CSCI 432 Handout 12: Shortest Paths in Linearly Interpolated Graphs

Name: ______________________________

2 April 2026

Problem 1 (SP in Linearly Interpolated Graphs)**.** *Given two weighted, directed graphs $G_0 = (V, E, \omega_0)$ and $G_1 = (V, E, \omega_1)$, define the linear interpolation at parameter $\tau \in [0, 1]$ to be*

$$G_\tau := (V, E, \omega_t := \tau\omega_1 + (1 - \tau)\omega_0).$$

Given a source $s \in V$ and a sink $\tau \in V$, we want to a data structure that will efficiently answer the query of: for $\tau \in [0, 1]$, what is the shortest path from s to t?

Recall Dijkstra's algorithm from our last class period. Here, we provide pseudocode for Dijkstra's SP algorithm for easy reference (this computes just the distance; can you adjust this to find the path too?):[1]

Algorithm 1 Dijkstra's Shortest-Path Algorithm

Input: A weighted, directed graph $G = (V, E, \omega)$ with non-negative edge weights; a source vertex $s \in V$
Output: Shortest-path distances $\mathrm{dist}[v]$ for all $v \in V$

Initialize dist, a data structure indexed by V, with value ∞ for all vertices except $\mathrm{dist}(s) = 0$
Q min-priority queue storing the elements off V, with $\mathrm{dist}[v]$ the priority of v for all v)
while $Q \neq \emptyset$ **do**
    $u \leftarrow \mathrm{pop}(Q)$
    for each (u, v) out of u **do**
        $alt \leftarrow \mathrm{dist}(u) + \omega(u, v)$
        if $alt < \mathrm{dist}(v)$ **then**
            $\mathrm{dist}[v] \leftarrow alt$
            DECREASE-KEY$(Q, v, \mathrm{dist}[v])$
        end if
    end for
end while
return dist

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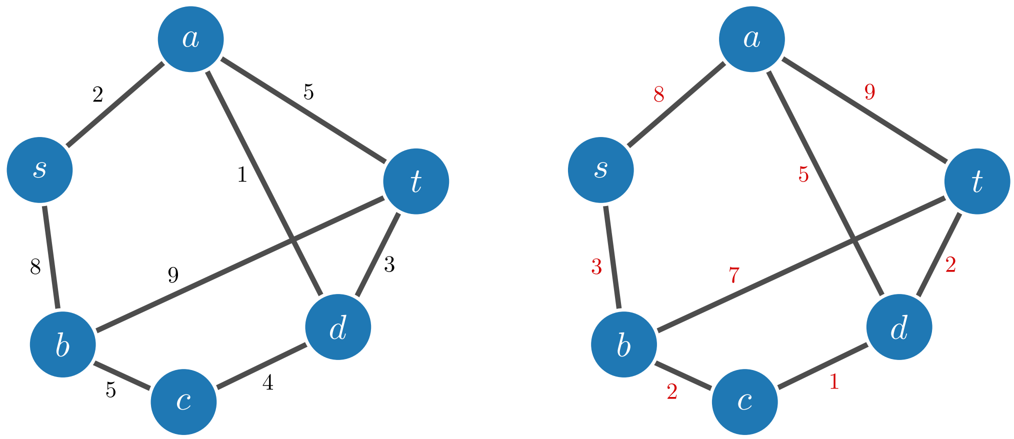
[1]Initial pseudocode was generated from CatChat, using the prompt "simple psuedocode for dijkstra in latex using the algorithmic package (not algpseudocode)"



(a) G_0 (b) G_1

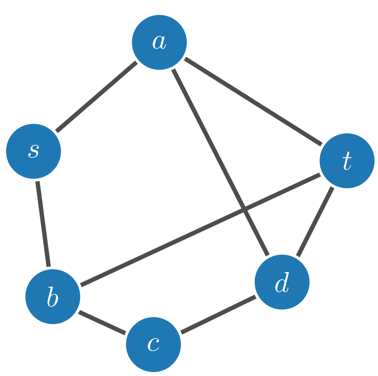
Figure 1: The starting and ending graphs that we linearly interpolate between. Here, let all edges be directed from left to right.

Understanding the Problem

1. For G_0 and G_1, use Dijkstra's algorithm to compute a shortest path from s to t in each graph (denote them by p_0 and p_1, respectively). What are their lengths (denote them by ℓ_0 and ℓ_1, respectively).

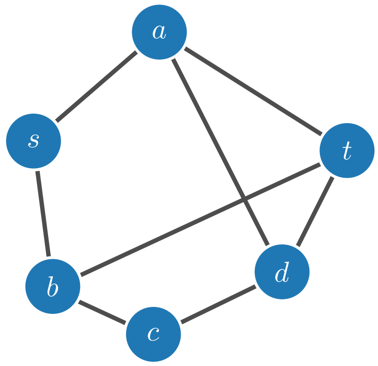
   Answer

2. Compute $G_{0.5}$ and find a shortest path from s to t. What is its length? (Denote the path and lengthy by $p_{0.5}$ and $\ell_{0.5}$).



   Answer

3. Compute $G_{0.25}$ and find the length of a shortest path from s to t. What is its length?



Answer

4. For which value of t are the lengths of p_0 and p_1 the same (Note: there is one value t for which this holds; we denote this by t_{same})?

Answer

5. What are the values of all paths between s and t at t_{same}. Is there any time that the shortest path is not p_0 or p_1? (Hint: look at the graphs of path length versus parameter τ for all paths from s to t).

Answer

6. Provide an example of two graphs for which there exists a t for which $p_t \notin \{p_0, p_1\}$.

Answer

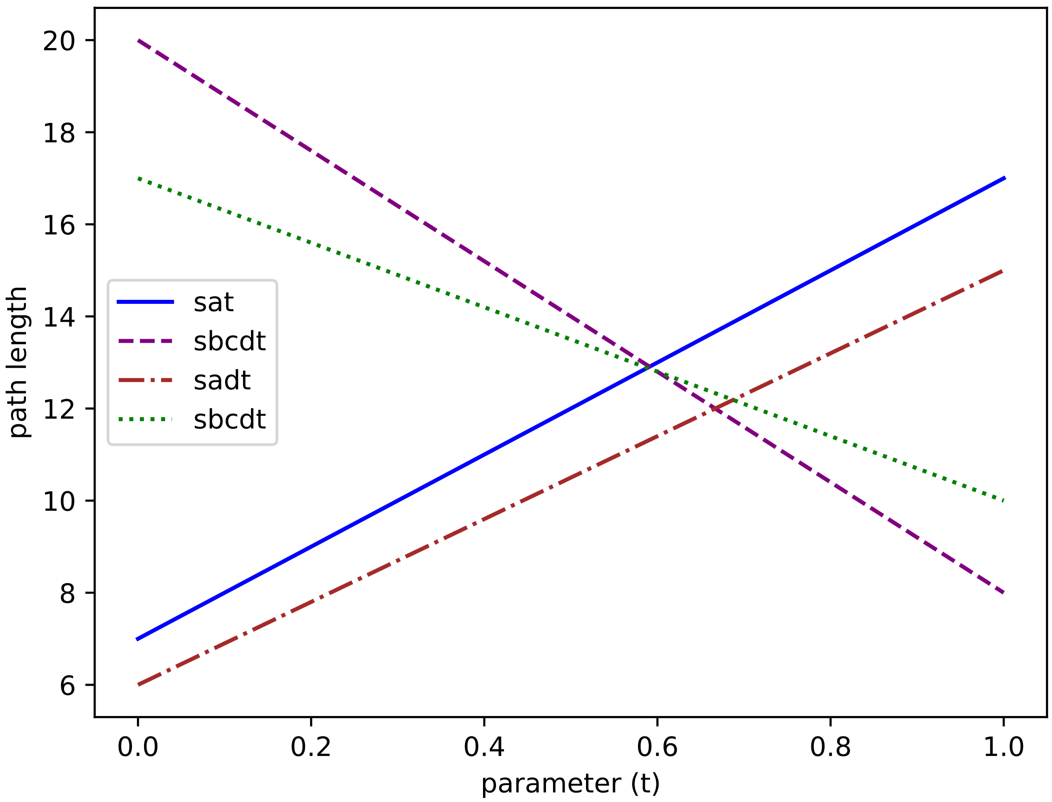
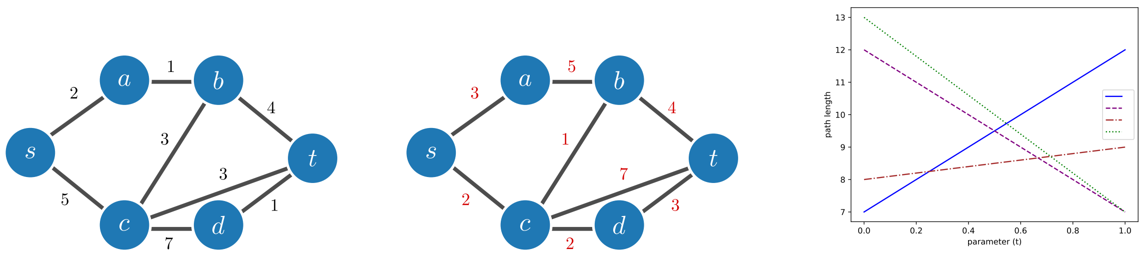


Figure 2: Path lengths.

The Algorithm

1. What are some observations we have about the plot above? When are the shortest paths not unique?

   Answer



(a) H_0 (b) H_1 (c) Path Lengths

Figure 3: The starting and ending graphs that we linearly interpolate between. Here, let all edges be directed from left to right.

2. Consider the linearly interpolated graph in Figure 3, along with the graph of the lengths of all paths from s to t. What are the shortest path representatives for each parameter τ?

   Answer